



DroneTalk

Natural Language Interface for
Drone Command and Control



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Our Team



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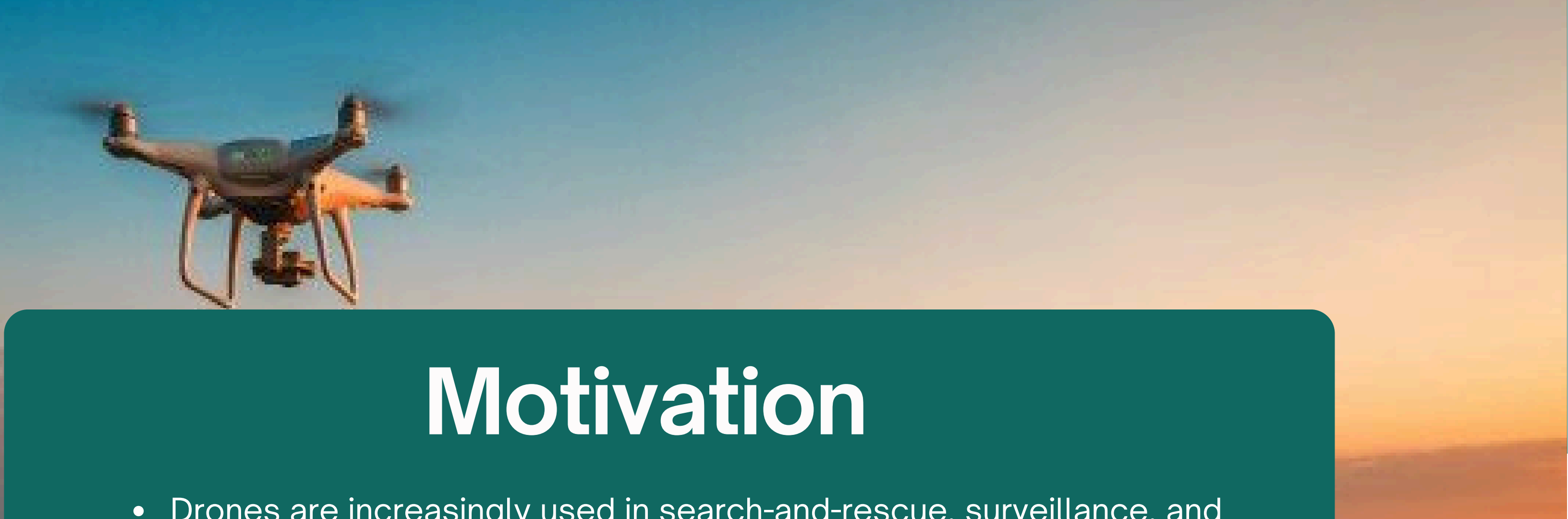
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Motivation

- Drones are increasingly used in search-and-rescue, surveillance, and forensic investigations.
- Current control methods rely on manual joysticks or predefined mission planning → not intuitive.
- Natural language control can make drones easier to operate and reduce response time.
- Our project explores how NLP can parse voice/text commands and convert them into structured, machine-readable actions.





INTRODUCTION

- Drones are widely used in surveillance, disaster management, and forensic investigations.
- Current control methods rely on manual joysticks or complex mission planning interfaces.
- These methods can be slow, unintuitive, and require expert training.
- Our project introduces an NLP-based system that allows users to control drones using natural language commands.
- Example:
“Fly 50 meters north and take 3 pictures.”
is automatically converted into structured commands the drone can be executed

Problem Statement

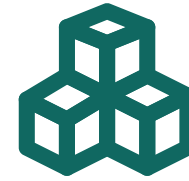
Controlling drones today relies mainly on manual joysticks or predefined mission scripts, which can be slow, unintuitive, and require technical expertise.

In critical situations like search and rescue, surveillance, or forensic investigations, operators need to react quickly — typing code or manually adjusting controls wastes time.



This bridges the gap between human language and machine-readable drone instructions, making drone operation faster, easier, and more accessible.

**Our Project Aims to
solve this by building
a system that can:**



Understand natural
language commands



Extract key information
such as intent (fly, capture)



Convert it into structured
drone commands

PipeLine

01



Command Input

User gives a voice or text command (e.g., “Fly 50 meters north and take 3 pictures”).

02



NLP Processing

- Intent Classification: Identify the main action (fly, capture, land, etc.).
- Slot Filling: Extract parameters (direction = north, distance = 50m, count = 3).

03



Command Generation

Convert parsed data into a structured command format (JSON or key-value pairs)

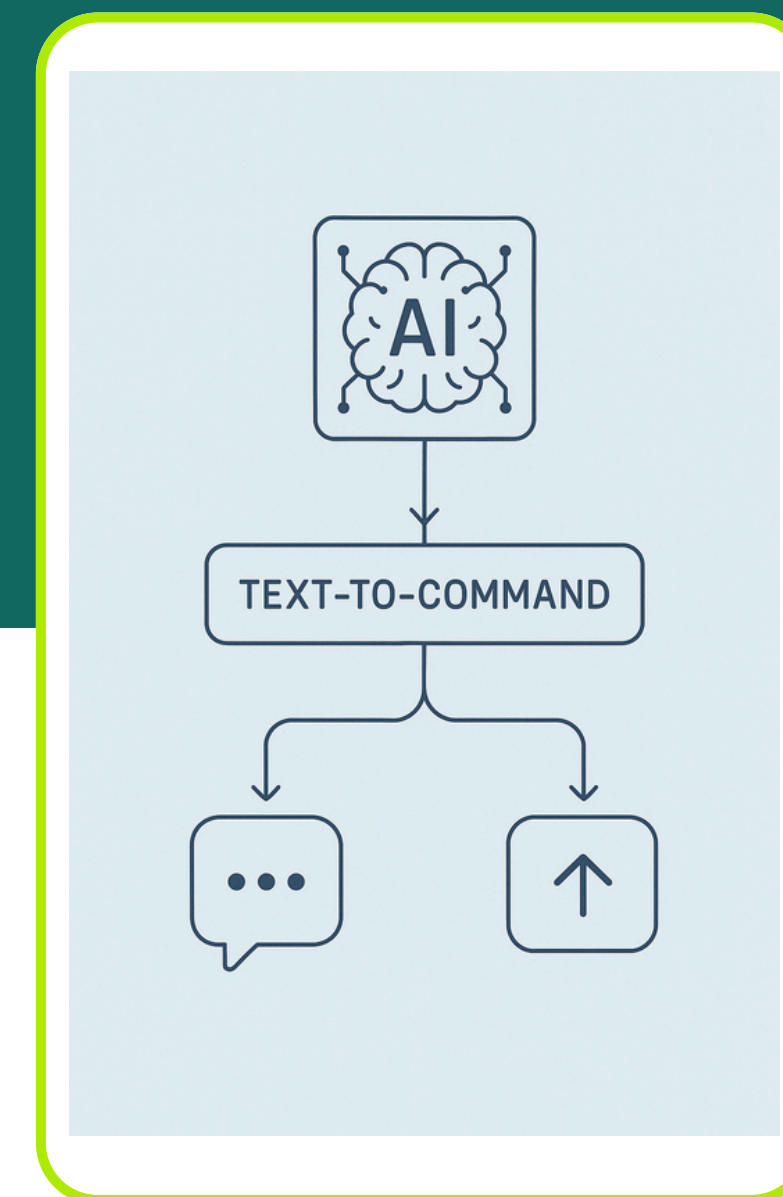
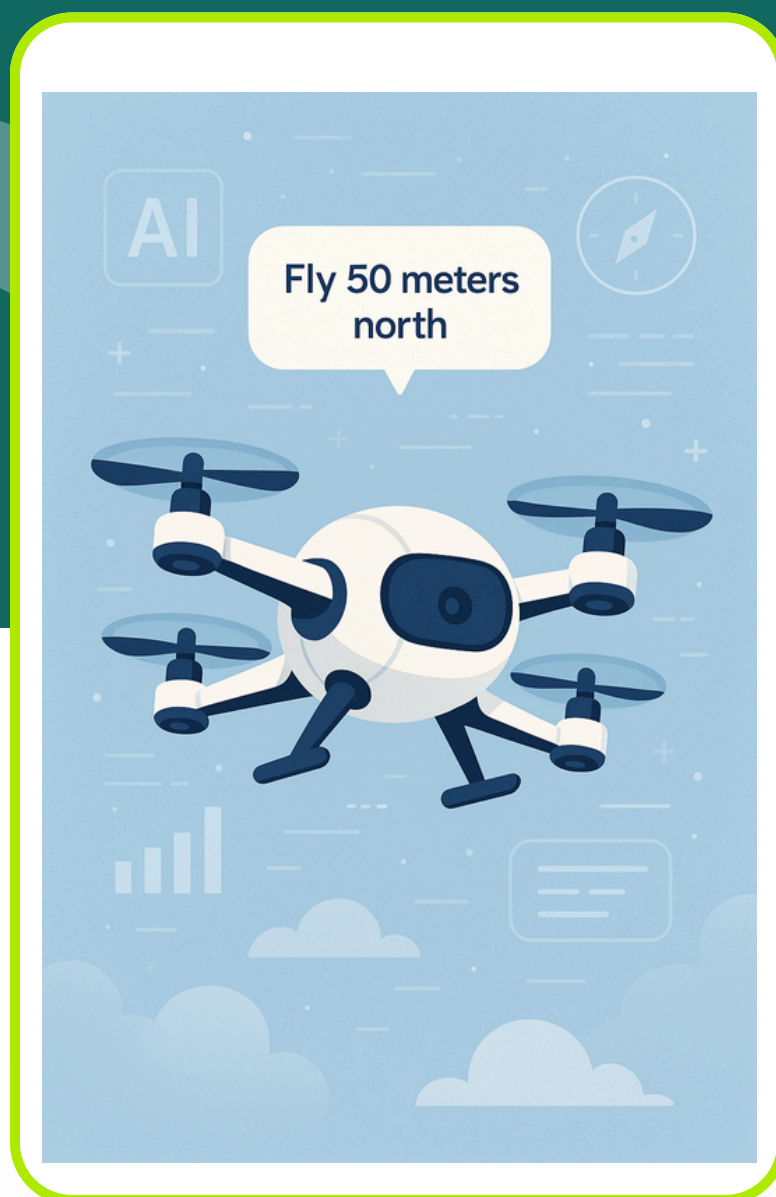
04



Execution/ Simulation

Pass the structured command to a drone simulator or display it as output.

A Visual Representation

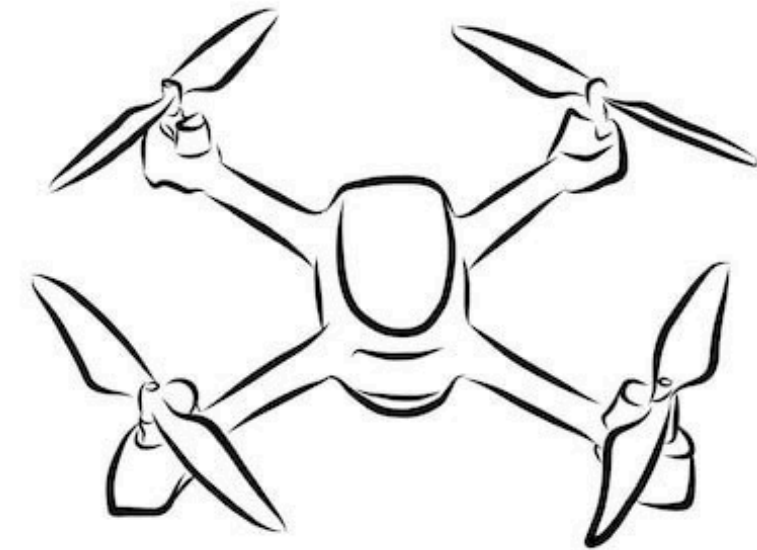


Project Timeline



Expected Outcome

- Working prototype that accepts text/voice commands
- Generates structured, machine-readable output for drone execution
- Potential for integration with real drone autopilot systems
- Future application: Search & rescue, disaster response, forensic investigations





Voice
Pilots



THANK YOU